

# Implementation of a tele-neurophysiology reporting network: our experience at ICS Maugeri

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## 01 BACKGROUND

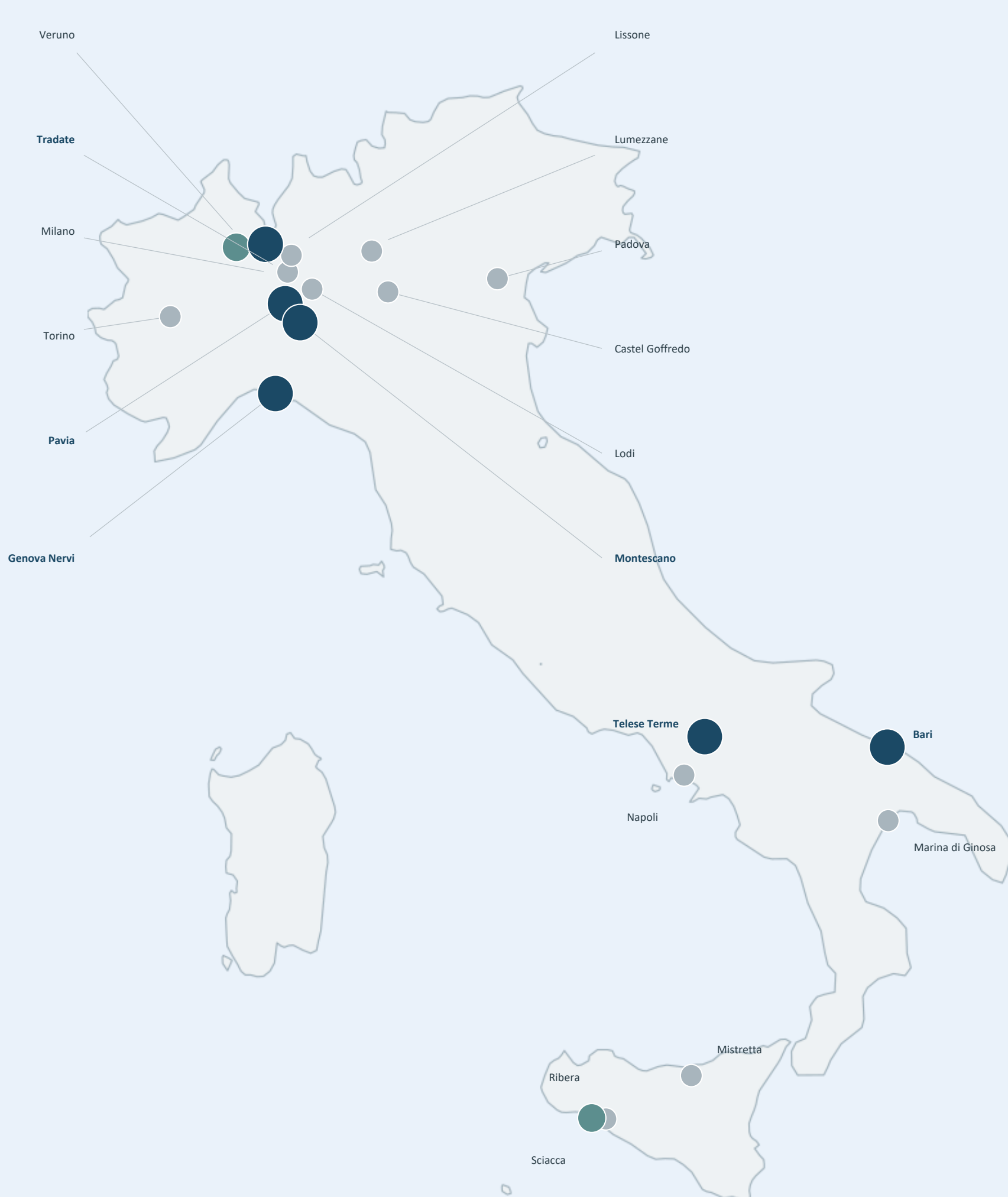
Demand for clinical neurophysiology is crucial in rehabilitation settings, where timely access to EEG, EMG and Motor and Sensory Evoked Potentials is often essential for diagnostic and prognostic decisions. This is especially relevant in Italy, where clinical neurophysiology is concentrated within neurology based training rather than a standalone pathway. We described our experience within ICS Maugeri in implementing a tele-reporting neurophysiology network.

## 02 METHODS

We performed a retrospective descriptive analysis of the network activity from 2018 to 2025. A shared digital infrastructure (EBneuro) enabled remote review, reporting, archiving, and digital signature across multiple Maugeri sites. Examinations included EEG, ENG/EMG, somatosensory evoked potentials, and motor evoked potentials.



## OUR NETWORK



**Fig. 2** — ICS Maugeri neurophysiology network across Italy. The map displays all centers equipped with neurophysiology devices and potentially connectable to the tele-reporting infrastructure. Colored sites represent centers that have actively joined the tele-neurophysiology network, in light green the ones which were formerly connected but later exited after the availability of an on site neurophysiologist.

## 03 RESULTS

A total of 1048 examinations were managed through the network. Main clinical indications included epilepsy follow-up, evaluation of neuropathies and myopathies, and assessment of central sensory or motor pathways in rehabilitation patients. The network allowed continuity of diagnostic activity and flexible redistribution of reporting workload between sites.



Remote review



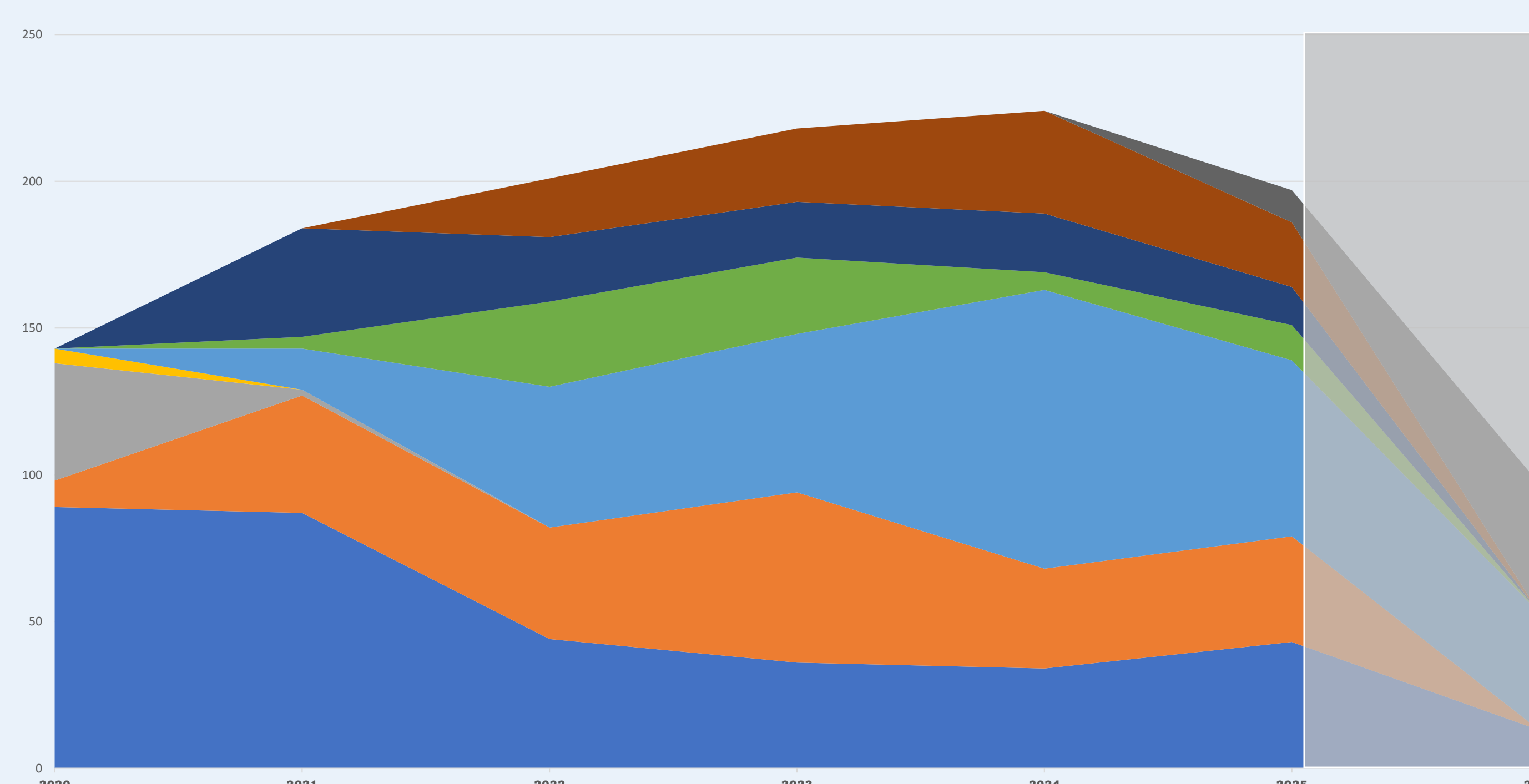
Centralized reporting



Digital archiving



Digital signature



### Area graphic

**Fig. 1** — Annual volume of tele-reported neurophysiology examinations across ICS Maugeri network. The graph shows the activity trend across participating sites from 2020 to 2026. The gray-shaded segment represents updated preliminary 2026 activity, increasing the overall number of remotely managed examinations to 1264.



### TEAM



## 04 CONCLUSIONS

A structured tele-neurophysiology model is feasible in a multicenter rehabilitation system and may improve access to specialist neurophysiology. Standardized procedures, staff training, and clear organizational governance are key for feasibility.

## 05 KEY POINTS

- 01 Multicenter Italian tele-neurophysiology experience
- 02 1048 examinations managed remotely (2018 – 2025)
- 03 Applicable to EEG, ENG/EMG, SEP and MEP workflows
- 04 Standardization is crucial for scale-up

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